

TinyImageNet Dataset Project

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Contents

Title	2
Abstract.....	2
About The Dataset	2
Related Work	2
Methodology.....	3
Transfer Learning Models	3
Few-Shot Learning Approaches	3
Results and Analysis.....	4
Training vs Validation Accuracy Curve for ResNet-50 (Adam).....	4
Training vs Validation Loss Curve for ResNet-50	4
Model-wise Accuracy Comparison Bar Plot	5
Few-Shot Accuracy Comparison (ProtoNet vs MAML)	6
MAML Meta Accuracy Progression over Iterations	7
t-SNE Visualization of Query Embeddings	7
Conclusion.....	8
Best Validation Accuracy of Each Model	8
References	10

Title

Transfer and Few-Shot Learning on Tiny ImageNet: A Comparative Study of Deep Architectures and Meta-Learning Approaches

Abstract

We conduct a comprehensive empirical study on various deep learning architectures and metalearning approaches on Tiny ImageNet. I started with baseline CNNs, transfer learning with ResNet50, MobileNet and EfficientNet variants. Later, I adopted a few shot learning paradigm such as Prototypical Networks, Siamese Networks and Model Agnostic Meta Learning. The best result was obtained with ResNet50 with full fine tuning of 52.71% of validation accuracy. With 60.13% meta accuracy, MAML performed better than few shot methods. I perform experiments that demonstrate the importance of model specific hyperparameter tuning and show the advantages of MAML in low data regimes.

About The Dataset

Given that Tiny ImageNet is a subset of ImageNet, and ImageNet is widely known as a difficult object classification benchmark with 1,000 classes and large image dimensions (224x224), Tiny ImageNet is quite challenging benchmark, especially with its relatively small images (64x64) and 200 classes. I hope to test standard transfer learning pipelines with recent few shot learning algorithms to find the best approach to this problem.

Related Work

These have been widely employed in transfer learning using large pretrained networks such as ResNet, MobileNet and EfficientNet. On the other hand, few shot learning aims to generalize from the very limited data using methods such as Prototypical Networks, Siamese Networks, and MAML. I investigate this synergy and benchmark these paradigms under consistent settings of operation.

I got the Dataset and Preprocessing from Tiny ImageNet dataset from Kaggle. It comprises 100,000 images across 200 categories. The images were resized to 64x64 and normalized. I generated five support and query sample per class (5-shot) for few shot learning.

Methodology

Transfer Learning Models

- ResNet-50 (Frozen Layers): The first 84 layers were frozen and a fully connected head of 512 neurons and 0.5 dropout was trained. Optimizer: SGD, LR=0.001. Validation accuracy peaked at 1.81%.
- ResNet-50 (Fine-tuned): All layers were unfrozen. One FC layer with 512 neurons. Optimizer: Adam, LR=0.0001. Achieved validation accuracy of 52.71% at epoch 26.
- MobileNet: Two FC layers of 1024 neurons each with 0.3 dropout. Optimizer: SGD, LR=0.001. Achieved 46.23% validation accuracy.
- EfficientNetB0: Single FC layer of 1024 neurons with 0.5 dropout. Adam, LR=0.0001. Max val accuracy: 17.60%.
- EfficientNetB3: Two FC layers with 1024 neurons each, 0.5 dropout. Adam, LR=1e-5. Achieved 46.76% val accuracy.

ReduceLROnPlateau and EarlyStopping were used in all models. We did extensive hyperparameter tuning through different dropout rates, learning rates, optimizers, and FC configurations.

Few-Shot Learning Approaches

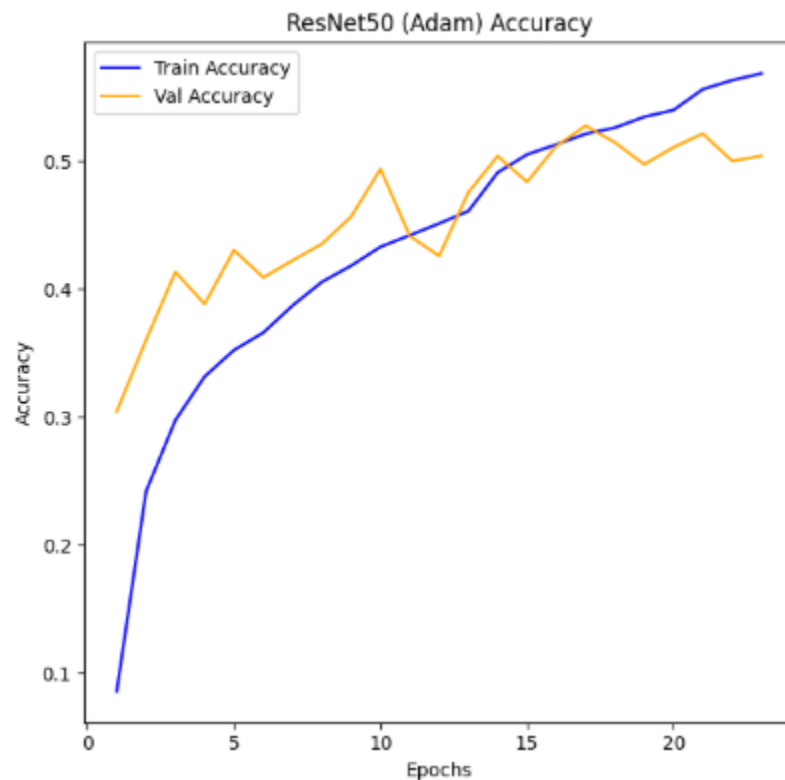
- Prototypical Networks: Given the support set embeddings, prototypes were computed per class. Euclidean distance to these prototypes was used for the classification. Achieved 43.36% accuracy.
- Contrastive loss is trained on ResNet-50 embeddings. But, but accuracies were unstable, had accuracies around 50% and even failed to predict a class at thresholds of 0.3.
- MAML: Implemented with inner learning rate of 0.04 and 5 adaptation steps. Obtained peak meta-accuracy of 60.13% at iteration 250. Shows strong generalization even in few-shot setups.

Results and Analysis

The following visuals summarize my findings:

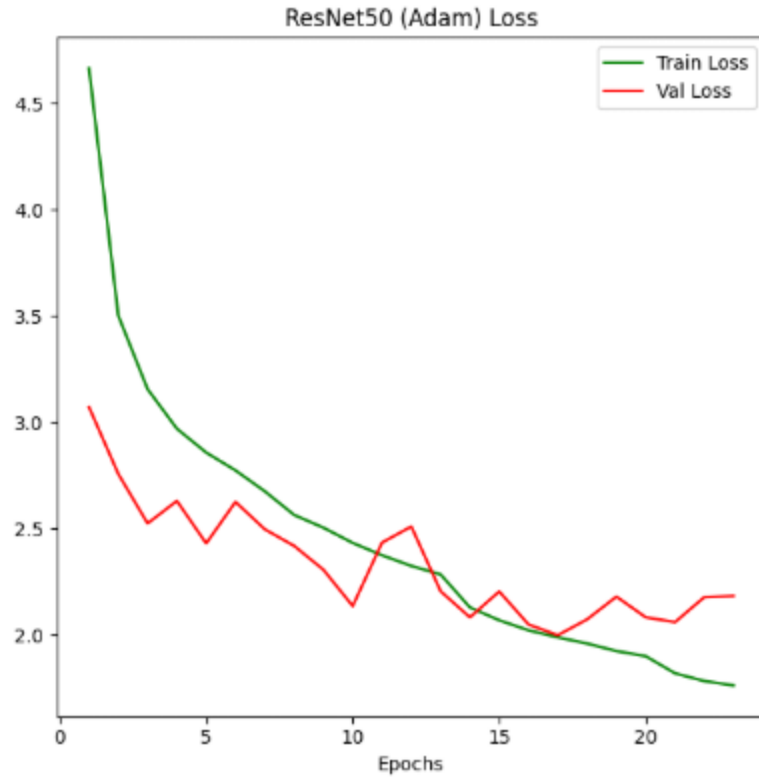
Training vs Validation Accuracy Curve for ResNet-50 (Adam)

The graphs display the training and validation accuracy. Both accuracy values improve as the training progresses, with the training accuracy reaching a higher value than the validation accuracy. This indicates that the model is generalizing well to the training data but could potentially benefit from further fine-tuning to improve validation performance.



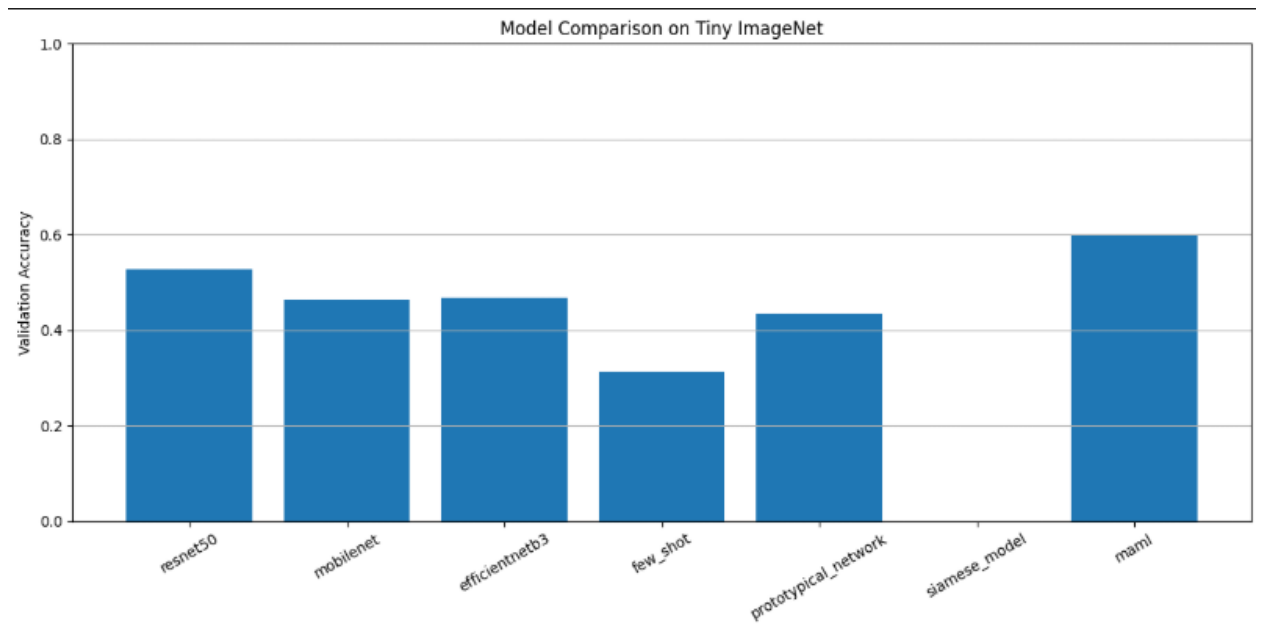
Training vs Validation Loss Curve for ResNet-50

The graphs shows the training and validation loss across epochs. The training loss decreases significantly, indicating that the model is learning effectively. However, the validation loss plateaus towards the later epochs, which suggests that the model may be starting to overfit, as it performs better on the training data than on unseen data.



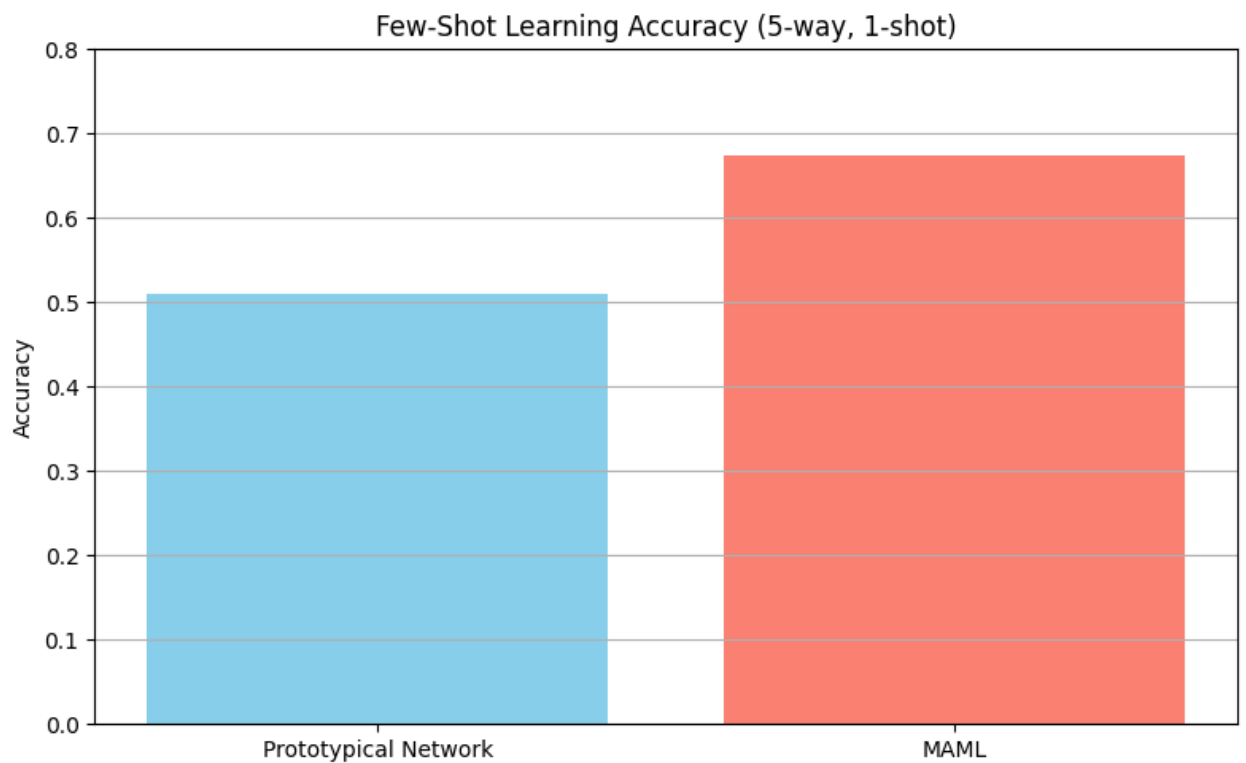
Model-wise Accuracy Comparison Bar Plot

The best validation accuracy of each model is plotted on this bar satisfy. In particularly, MAML is at 60.13%, followed by ResNet-50 (Adam), demonstrating the power of meta learning in few shot situation.



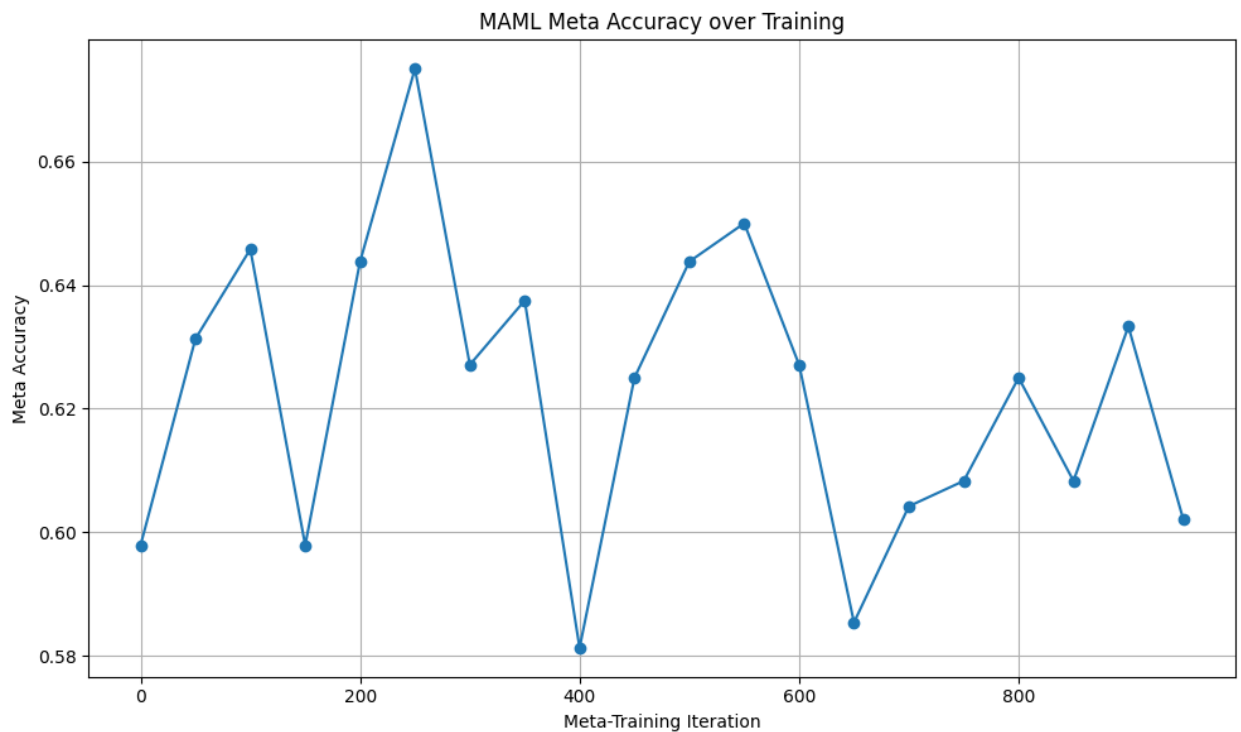
Few-Shot Accuracy Comparison (ProtoNet vs MAML)

The bar chart compares accuracy of few-shot approaches. Interestingly, MAML is much better adapted to quickly learn from little data compared to Prototypical Networks.



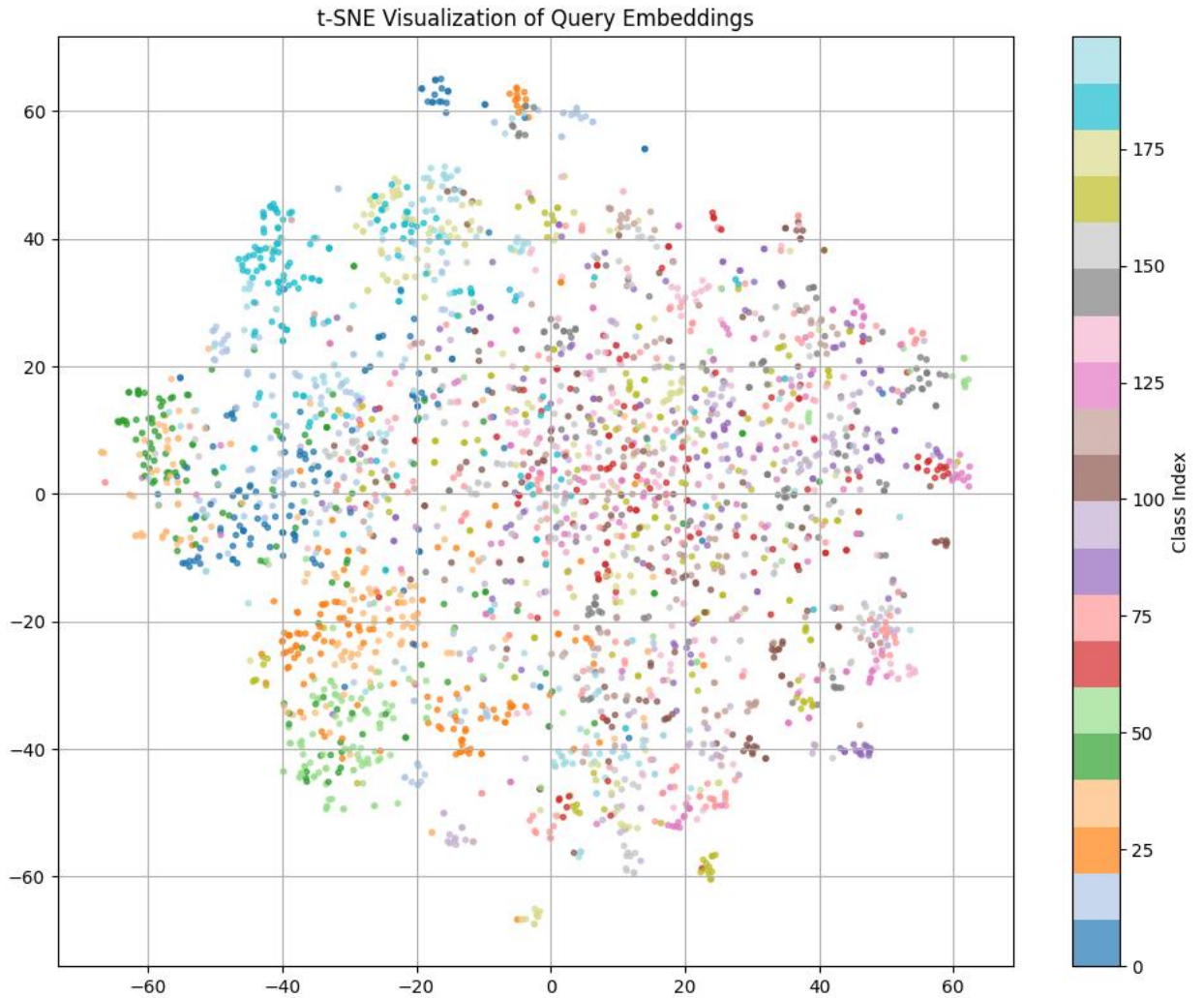
MAML Meta Accuracy Progression over Iterations

Plotted is the meta accuracy of MAML for 1000 iterations of training. Overall, there are some fluctuations but the trend is consistently strong with multiple peaks above 64%.



t-SNE Visualization of Query Embeddings

High dimensional query embeddings are projected into 2D using t-SNE plot. For many classes, clustering patterns are good (each dot is an image embedding colored by class index).



Conclusion

I conclude that fine tuned ResNet-50 is best in traditional transfer learning on Tiny ImageNet. MAML is much more effective than other approaches in fewshot settings. These findings are evidence of the power of full fine tuning for transfer learning and also as the promise of meta learning for data sparsity. There is future work exploring hybrid models that combine meta-learning with transfer based priors.

Best Validation Accuracy of Each Model

ResNet-50 **52.72%**

MobileNet 46.24%

EfficientNetB3	46.76%
Prototypical Net	43.36%
MAML	60.13%
Siamese Network	0.00%
Few-Shot (Baseline)	31.28%

References

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